

Tudor Andrei Diac

Lund, Sweden | Email | +40 740 273 848 | Google Scholar | GitHub

Research Interests

Machine learning grounded in computational mathematics, numerical methods, and statistical modelling, with an emphasis on small, interpretable, and calibrated models and on quantifying what learned representations actually capture; representation and latent-prediction methods for dynamical and physical systems; efficient, data-centric pipelines and edge-ready deployment of ML systems.

Education

Lund University, MSc in Computational Science (Scientific Computing) Sep 2024 – June 2026

- **MSc Thesis:** *Pretraining, Quantization, and Evaluation of Romanian Small Language Models*. Supervisors: Anders Björkelund (LU) **Grade: VG (Pass with distinction)**.

University of Bucharest, Bachelor in Physics Informatics Oct 2021 – July 2024

- **Bachelor's Thesis:** *The Study of the Production of Charm Mesons and Tau Neutrinos in the DsTau Experiment* (CERN collaboration). Supervisors: Oana Ristea (UB, Faculty of Physics), Elena Firtu (ISS-INFLPR), Titi Preda (ISS-INFLPR). **Grade: 10/10**.

Research Experience

Research Collaborator, ISS - Institute of Space Science, Romania May 2025 – Ongoing

- Contributing to interdisciplinary research in Statistics, Scientific Computing, and NLP, with a focus on low-resource language model evaluation and authorship verification using frequency and structural metrics
- Leading a research project on NLP with a focus on domain-specific Romanian language models, including crawling, preprocessing, and transformer-based training

Research Technician III - DsTau(NA65) CERN Project, ISS - Institute of Space Science, Romania April 2024 – Sep 2024

- Introduced AI/ML to the workflow by prototyping a particle-momentum estimation algorithm, integrating model outputs into the DsTau (NA65) pipeline
- Built an ensemble of neural networks to discover $D_s \rightarrow \tau$ candidate decays; achieved 94.2% accuracy on a held-out test set, shifting detection from manual to automated screening for the collaboration

Student Internship, CERN, Switzerland Nov 2023 – Dec 2023

- Participated in advanced Theoretical Particle Physics courses and academic seminars during a focused internship program at CERN

Publications

T.A. Diac, A.F. Neagoe, M.C. Raportaru, A. Tolea, R.-M. Drăgan, A. Nicolin-Żaczek
Statistics of a large-scale Romanian corpus for language modelling.

Romanian Journal of Physics (RJP), 2025. DOI: <https://doi.org/10.59277/RomJPhys.2025.70.111>

T.A. Diac, P.F. de Viana, A.F. Neagoe, A. Oprea, M.C. Raportaru, A. Nicolin-Żaczek
Edge-Ready Romanian Language Models: Training, Quantization, and Deployment.

AI (MDPI), 2026. DOI: <https://doi.org/10.3390/ai7020061>

A. Nicolin-Żaczek, P.-A. Gogîta, T.-G. Dumitru, F.-I. Constantin, **T.A. Diac**, A. Neagoe, M.-C. Raportaru
Scale-free to Pareto-Tsallis transitions in the distributions of waiting times: weather, sea-level, currency trading and automotive datasets.

Journal of Physics: Complexity, 2026. DOI: <https://doi.org/10.1088/2632-072X/ae8aa5>

Conferences

Tudor-Andrei Diac, Mihaela Carina Raportaru, Alexandru Nicolin-Żaczek.

Training Large Language Models on a Novel Romanian Corpus.

Poster presented at the **International Balkan Workshop on Applied Physics and Materials Science (IBWAP)**, Constanța, Romania, July 2025.

Research Projects

Self-Supervised Representation Learning for Non-Invasive Brain Decoding

Ongoing

- Built a modality-invariant JEPA (spatio-temporal masking, FNO predictor, EMA target encoder, VICReg, gradient-reversal domain adversaries) mapping 306-ch MEG and 64-ch EEG through per-modality anatomical adapters into one shared transformer trunk; evaluated on Nieto inner-speech and Meta's Brain2Qwerty (264GB, 35 subjects)
- Built the diagnostic scripts that made the model's behaviour visible: shuffled-context checks on the pretraining loss, 20-permutation null distributions, confound floors, and person-level de-duplication that caught a subject-ID leak in the released dataset (the same participant appearing under several IDs)
- Showed the standard variance-collapse detector used in this family of models is computed on the wrong axis, so it reports healthy statistics on an encoder producing identical outputs for every input; this was reproduced on two datasets and two architectures

Embedded Sensing & Edge AI for Drones

Completed

- Developed a Jetson Nano/Orin NX stack interfaced with Arduino sensors (temperature, gas, sound, motion) for real-time telemetry and anomaly detection.
- Built a terminal UI with live plots, status panels, and alerting; logged streams to InfluxDB; implemented threshold & variability detectors for autonomous event flags.
- Focus: robust serial comms, low-latency pipelines, and on-device inference for bandwidth-constrained deployments.

Technical Skills

Math/Stats - Probability and Statistics, Markov Processes, Monte Carlo, Extreme-value modeling, Linear Algebra; scientific computing workflows

Programming - Python, C/C++, Bash, Git & GitHub

AI/ML/NLP - Tensorflow, HuggingFace (Transformers/Datasets); CNNs, RNNs, LSTMs, GRUs, and Attention-based models; tokenizer design; evaluation; quantization, llama.cpp/Ollama

Systems and Acceleration - Linux, profiling, data pipelines (JSONL/packing), reproducible training configs

Embedded/Edge - NVIDIA Jetson (Nano/Orin NX), Raspberry Pi, Arduino, serial comms, InfluxDB, real-time telemetry UIs

Extra - Solving Competitive Coding Problems (Over 300 C/C++ problems on <https://www.pbinfo.ro/>)

Accomplishments

- Selected as **1 of 12 students nationwide** (across Romanian Physics programs) for a **CERN internship** with full tuition and expenses covered
- **LLM-jp Collaboration member** (<https://llm-jp.nii.ac.jp/en/>)